

Pascal Grange

📍 Bordeaux, France [in pgrange1](#) [G pgrange](#)

About Me

Senior Software Engineer with expertise in security, Bitcoin, and Cardano. Passionate about building resilient, scalable systems and leading teams with a lean management approach. Experienced in financial infrastructure, Layer 2 solutions, and open-source contributions. Pragmatic, hands-on, and driven by continuous learning and innovation.

Education

PhD **Université de Bordeaux**, Computer Science Bordeaux, France
Distributed Systems: Transparency, Hiding, and Associated Tools 2005

Experience

French National Cybersecurity Agency (ANSSI), Chief Information Security Officer Paris, France
(CISO) Consultant for the Innovation Lab Dec 2024 – present

21research, Founder Bordeaux, France
• Building, securing and hosting a Bitcoin Lightning node Dec 2023 – present
• Exploring the development of a [Lightning Hardware Wallet](#) (Kotlin)
• Developing LN paid services (Rust, Elm)

IOG, Senior Software Engineer Fully Remote
• Developing [Hydra](#), the layer 2 solution for the [Cardano](#) blockchain (Haskell) Sept 2022 – Sept 2023

Sinch, Senior Software Engineer Bordeaux, France
Expanding a local SaaS product to a global worldwide solution May 2020 – Aug 2022
• Migrating from baremetal to the cloud (AWS, Terraform, Packer, Ansible)
• Handling local regulations, security and total cost of ownership (ISO-27001, GDPR)
• Implementing 24/7 on-call duty contract to cover all timezones
• Facilitate post-incident reviews

LGO, Chief Information Security Officer (CISO) Bordeaux, France
Building and securing the leading (at the time) institutional Bitcoin exchange in Europe. Mar 2018 – May 2020
Approx. \$500.000.000 equivalent traded on the platform until february 2020.
• Building an architecture targeting several millions orders per second (Aeron Cluster, Java)
• Test driven infrastructure as code (Terraform, Packer, Ansible, role-unit)
• Automated failure management (GCloud, disk expansion, service restart, machine restart)
• Implementation of an on-call contract (just in case)

Orange Business Services, Manager of expert teams in web, telephony and NFC Bordeaux, France
Supporting change to evolve from a culture where system administration is performed Oct 2011 – May 2018
manually to a *devops* and *infrastructure as code* culture. In charge of designing, costing and building several customer platforms. Responsible for their quality of service, security and profitability for the company
• Management of a team of systems and network engineers and project managers (agile software development, lean-thinking, standards, technical sharing workshops, recruitment)
• Management of security audits, formalization of responses and follow-ups
• In charge of a multi-country web authentication platform using smartphone — *Mobile Connect*

- In charge of shared cryptography service — *Hardware Security Module as a Service* (Thales ncipher HSM)
- In charge of an e-commerce website — client annual income over 150 M€ (GitLab, Ansible, HA-proxy, Varnish, MySQL)
- In charge of an NFC mobile payment platform (Mastercard certification, SIM OTA Management, Thales Payshield HSM, Mobile Carrier networks interconnection)
- In charge of a Telephony platform with active replication over three distinct geographical areas - 100 000 calls a day (OMS Interactive Voice Response)

Orange Business Services, Software Developer

Bordeaux, France
Nov 2005 – Sept 2011

- Test Driven Development (Java, Javacard, Javascript, Python)
- frequent writing of executable specifications (FitNesse)
- building several continuous integration platforms (GitLab CI, Jenkins, Sonar, Cruise Control)
- team leadership, project management and writing of commercial offers (Scrum Master)

Univerté de Bordeaux, ENSEIRB, Teacher in higher education

Bordeaux, France
Sept 1999 – Aug 2005

Projects

bash_unit [↗](#)

Unit testing framework for Bash

- 600+ stars on GitHub, widely adopted in the community
- Included in Debian Sid, NixOS, Homebrew and ArchLinux, making it easily accessible for developers
- Enables test-driven development (TDD) for shell scripts with a simple syntax

soap-dust [↗](#)

Java soap library (discontinued)

- SOAP client and server library, including test framework
- Fits within a single jar file, needing no external dependencies

Skills

Programming: Bash, Java, JavaScript, TypeScript, Haskell, Rust, Elm, Python, Go

Operations: AWS, GCP, Terraform, Packer, Ansible, Docker, Nomad, Consul, Vault, Prometheus, OpenTelemetry, CI/CD

Security: Risk assessment, pentesting, hardware security, security policy design, bug bounty programs, cryptography, compliance (ISO 27001, RGPD, PSAN)

Management: Lean management, TDD culture, team coaching, scaling engineering teams, mentoring, fostering knowledge sharing, security awareness training

Bitcoin & Blockchain: Bitcoin protocol, Lightning Network, OpenTimeStamps, Hydra (Cardano L2), secure custody solutions

Languages: English: Fluent / French: Native / German: Basic, eager to improve